FlytBase UAV Systems Engineer Assessment

Multi-Drone Coordination System

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Summary:

This document presents the **Multi-Drone Coordination System (MDCS)** for the FlytBase testing facility — 16 drone docks arranged in a 4×4 grid with 2–3m spacing. The system draws inspiration from aviation Air Traffic Control (ATC) principles and adapts them to the specific geometry and operational needs of a dense, co-located drone fleet.

The core safety strategy is **three-dimensional geometric deconfliction**: each drone row is permanently assigned to a unique horizontal transit altitude, and each column is assigned as a dedicated vertical corridor. This eliminates path intersections during horizontal transit — the highest collision-risk phase — without requiring complex real-time AI negotiation. All 16 drones can operate simultaneously under this system.

KEY INNOVATION	Row-altitude + Column-corridor deconfliction enables all 16 drones to fly simultaneously while mathematically eliminating horizontal path conflicts. The rules are geometric and deterministic — no AI required for basic deconfliction.

1.Layout and Core Architecture

The 4×4 Dock Grid

Docks D1–D16 are arranged in row-major order. Row 1 = D1–D4 (north), Row 4 = D13–D16 (south). Columns run 1–4 from west to east. Each dock is 2–3m from adjacent docks.

	Column 1 (Corridor A)	**Column 2 (Corridor B)**	**Column 3 (Corridor C)**	**Column 4 (Corridor D)**	
Row 1	**D1 (Row 1)**	**D2 (Row 1)**	**D3 (Row 1)**	**D4 (Row 1)**	Transit Alt: 40m
Row 2	**D5 (Row 2)**	**D6 (Row 2)**	**D7 (Row 2)**	**D8 (Row 2)**	Transit Alt: 35m
Row 3	**D9 (Row 3)**	**D10 (Row 3)**	**D11 (Row 3)**	**D12 (Row 3)**	Transit Alt: 30m
Row 4	**D13 (Row 4)**	**D14 (Row 4)**	**D15 (Row 4)**	**D16 (Row 4)**	Transit Alt: 25m

Fig. 1 — 4×4 Dock Grid with Row-Altitude and Column-Corridor assignments.

2.Core Deconfliction Principle

The fundamental challenge is that 16 docks at 2–3m spacing create a high-density airspace where horizontal transit paths from different docks will inevitably cross. The MDCS resolves this through a two-axis assignment system:

- Row -Transit Altitude: each row is permanently assigned a unique horizontal flight altitude. Drones at different altitudes cannot collide horizontally regardless of heading.
- Column -Vertical Corridor: Each column has a designated lateral position for climbing and descending. Only one drone per column may occupy the 0–15m corridor at a time.

2.Altitude Layer System

Each row is permanently assigned at a transit altitude. These assignments are fixed and never reassigned during normal operations. The 5m separation between layers provides adequate vertical buffer for GPS altitude accuracy (±1.5m typical) plus a safety margin.

Row	Transit Altitude	Columns	Notes
Row 1 (D1–D4)	40 m AGL	Column 1–4	Northernmost row, highest altitude
Row 2 (D5–D8)	35 m AGL	Column 1–4	5m below Row 1
Row 3 (D9–D12)	30 m AGL	Column 1–4	5m below Row 2
Row 4 (D13–D16)	25 m AGL	Column 1–4	Southernmost row, lowest transit alt

SOP RULE 1.1	A drone MUST reach its row-assigned transit altitude before initiating ANY horizontal movement. Horizontal flight at non-assigned altitude is a safety violation requiring immediate RTH command.

Column Corridor Assignments

Column	Docks	Corridor Offset	Description
Column 1	D1, D5, D9, D13	0.0m lateral offset	True North corridor
Column 2	D2, D6, D10, D14	+0.5m East offset	Slight east shift
Column 3	D3, D7, D11, D15	+1.0m East offset	Further east shift
Column 4	D4, D8, D12, D16	+1.5m East offset	Easternmost corridor

3.Flight Phase Deconfliction

FLIGHT PHASE	ALTITUDE BAND	RULES / NOTES
GROUND / DOCKED	0 m	Charging, pre-flight checks
VERTICAL TAKEOFF	0 to 15 m	Column corridor — 1 drone/column max
TRANSIT CLIMB	15 to Row Alt	Climb to assigned row altitude

HORIZONTAL TRANSIT	Row-Assigned Alt	Row 1: 40m | Row 2: 35m | Row 3: 30m | Row 4: 25m
TARGET OPERATIONS	> 50 m or target alt	Payload, survey, inspection — unrestricted
RETURN TRANSIT	Row-Assigned Alt	Same altitude maintained — return to column entry
VERTICAL DESCENT	Row Alt □ 0 m	Column corridor — await clearance if occupied
DOCKED & CHARGING	0 m	Transmit 'DOCKED' status — corridor released

Fig. 2 — Complete altitude lifecycle for a drone mission from dock to target and back.

4.Standard Operating Procedures

SOP-TKO: Takeoff Procedure

1. Pre-flight check: battery >30%, GPS fix confirmed, comm link verified, flight plan filed with MDCS.
2. Request takeoff clearance from MDCS (automated or operator-approved).
3. MDCS checks: column corridor clear (0–15m), row transit altitude available, no conflicting RTH in progress.
4. On clearance: arm motors. Begin vertical ascent within 1m horizontal dock center at 3 m/s.
5. Climb to 15m (transit base altitude) within the assigned column corridor.
6. Transition to column offset position; continue to climb to row-assigned transit altitude.
7. Broadcast 'TRANSIT READY' to MDCS. Begin horizontal navigation to target waypoints.

SOP-TKO-1	ONLY ONE drone per column may occupy the 0–15m vertical corridor simultaneously. A requesting drone holds on ground (rotors idle) until the corridor is clear.

SOP-LND: Landing Procedure

1. Drone broadcasts RTH intent to MDCS with ETA for 60 seconds before initiating return.
2. MDCS assigns landing slots and reserves the drone column corridor.
3. Drone descends to row at transit altitude, navigates horizontally to column corridor entry point.
4. Broadcast 'CORRIDOR ENTRY'. Begin vertical descent within column corridor at 2 m/s.
5. Reduce to 1 m/s below 5m AGL for precision dock approach.
6. Land, disarm motors, transmit 'DOCKED' status. The corridor is released immediately.

SOP-LND-1	Battery < 15% triggers Priority Landing (P2). MDCS clears the drone column corridor immediately. All conflicting drones in that corridor hold or divert to hover.

SOP-TRN: Horizontal Transit Rules

- Drones must maintain row-assigned altitude ± 2m tolerance during all horizontal flights.
- Minimum horizontal separation between same-altitude drones: 10m at all times.
- Cross-row transit (drone needs to fly laterally through another row of altitude band) requires explicit MDCS clearance and temporary altitude reassignment.
- Maximum horizontal speed over the dock grid area: 5 m/s.
- Drones always yield to drones with a lower battery percentage when altitude bands are temporarily shared.
- Drones must broadcast position at 2 Hz during all transit phases.

5.Priority Rules and Conflict Resolution

When two drones have conflicting corridor or altitude needs, priority is determined by the following permanent hierarchy. Higher priority always overrides lower priority.

LEVEL	NAME	TRIGGER CONDITION	SYSTEM RESPONSE
P1	**CRITICAL — EMERGENCY**	Drone malfunction / comm loss / imminent collision	ALL drones hold; MDCS clears full path within 5s
P2	**URGENT — LOW BATTERY**	Battery < 15% - immediate corridor clearance	Column corridor reserved; conflicting drones hover or divert
P3	**HIGH — TIME CRITICAL**	Payload with hard deadline / regulatory window	Next available corridor slot assigned immediately
P4	**NORMAL — ROUTINE**	Standard mission, FIFO queued	Joins scheduling queue; takeoff when corridor free
P5	**LOW — REPOSITIONING**	Test / repositioning / non-urgent; preemptable	Last in queue; mission paused if P1–P3 needs resources

Fig. 3 — Priority levels P1–P5 governing conflict resolution between drones.

Emergency Protocols

E1 — Communication Loss

1. Drone immediately enters an autonomous hover for 10 seconds at the current altitude.
2. If comms not restored: execute pre-programmed RTH using the last valid route stored onboard.
3. MDCS flags the drone as 'COMMS LOST' and clears its predicted corridor path.
4. All other drones maintain a 20m horizontal buffer around the last known position of the comms-lost drone.
5. Ground crew alerted for visual monitoring if available.

E2 — Drone Malfunction (Partial)

1. Drone broadcasts MAYDAY code to MDCS via telemetry.
2. MDCS immediately identify the nearest safe landing zone (empty dock or designated emergency pad).
3. All drones within 50m altitude band are advised to diverge 30m laterally.
4. Malfunctioning drone prioritizes landing above all other mission objectives.
5. Post-landing: drone grounded pending engineering inspection.

E3 — Mid-Air Conflict Detection

1. If telemetry detects two drones within 8m horizontal separation at the same altitude layer: automatic diverge maneuver triggers — one ascends +5m, the other descends □5m based on priority.
2. MDCS logs into incidents as a safety event requiring mandatory post-flight investigation.
3. Both drones are grounded after mission completion for audit.

E4 — Total MDCS Software Failure

1. All airborne drones immediately reduce speed to 2 m/s and begin broadcasting position at 4 Hz.
2. Each drone executes an autonomous RTH home dock using onboard GPS navigation.
3. RTH is staggered: each drone waits (dock number × 5) seconds before descending, preventing simultaneous corridor conflicts. (D1 waits 5s, D2 waits 10s, ..., D16 waits 80s)
4. Manual override operator assumes control from backup console.
5. MDCS restart is attempted in parallel; operations resume only after full system health check.

7.KPI'S

KPI	Target	Measurement	Category
Minimum Separation Distance (MSD)	> 8m at all times	Telemetry log analysis post-flight	Safety
Corridor Conflict Rate	< 0.01 per 100 missions	MDCS incident log	Safety
Emergency RTH Success Rate	> 99.5%	Simulated drills + live tests	Safety

KPI	Target	Measurement	Category
System Uptime (MDCS)	> 99.9%	Server monitoring dashboard	Reliability
Mission Throughput	> 8 missions/hr (8 drones)	MDCS mission counter	Efficiency
Avg Corridor Wait Time	< 90 seconds	Queue timestamp logs	Efficiency
Corridor Utilization	> 70%	Time-in-use / time-available	Efficiency
Battery Incident Rate	0 losses to battery	Battery telemetry + landing logs	Safety

8.Scalabilty and Design Perspective

The MDCS is designed to scale beyond the current 16-dock configuration:

- Grid expansion: adding rows simply adds new altitude layers (5m increments below the existing lowest layer). No existing rules need to be changed.
- Column expansion: additional columns add new vertical corridors with lateral offsets. The column corridor locking rule is unchanged.
- Multi-site: multiple grids can operate independently under separate MDCS instances with airspace boundaries defined by geofences.
- Increasing drone count per dock: if dock technology allows multiple drones per dock, the corridor's time-slot system can be extended to manage multiple drones per corridor.
- Integration with UTM (Unmanned Traffic Management): the altitude layer and corridor framework map directly to UTM U-Space corridors and can be registered as formal airspace structures.